# Threshold Output Preservation under Bounded Score Perturbations

Research Architecture Runtime
`operator/thresholding`

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**Abstract**

Scalar threshold $A_T(x) = \mathbf{1}\{x \geq T\}$ is invariant outside the $2\varepsilon$ unstable band when $|x' - x| \leq \varepsilon$. Lean `LEAN_FULL`.

**This theorem is:** `Primitive`.

## 1 Problem

The operator returns the binary indicator $A_T(x) = 1$ iff $x \geq T$ (equality passes).

## 2 Stability notion

Perturbation: $|x' - x| \leq \varepsilon$. Pass buffer $x \geq T + \varepsilon \Rightarrow A_T(x') = 1$; fail buffer $x < T - \varepsilon \Rightarrow A_T(x') = 0$.

## 3 Definitions

**Definition 1.** $A_T(x) \Leftrightarrow x \geq T$.

**Assumptions.** $\varepsilon \geq 0$ and $|x' - x| \leq \varepsilon$.

## 4 Theorem

**Theorem 2.** *If $|x' - x| \leq \varepsilon$, then $x \geq T + \varepsilon \Rightarrow x' \geq T$ and $x < T - \varepsilon \Rightarrow x' < T$.*

## 5 Intuition

Moving $x$ by at most $\varepsilon$ cannot cross $T$ from outside the $\varepsilon$-neighborhood.

## 6 Examples

**Example 3.** *$T = 3$, $x = 5$, $\varepsilon = 0.2$: $x \geq T + \varepsilon$, so every $x'$ in the ball still passes.*

## 7 Proof

From $|x' - x| \leq \varepsilon$ one has $x' - \varepsilon \leq x \leq x' + \varepsilon$. If $x \geq T + \varepsilon$ then $x' \geq x - \varepsilon \geq T$. If $x < T - \varepsilon$ then $x' \leq x + \varepsilon < T$ (Theorem 2).

## 8 Formal statement

Lean module: `Research.Operators.Threshold.Preservation`. Source file: `lean/Research/Operators/Thresho`
Theorems: `threshold_preservation`. Certificate: `lean/certificates/thresholding/threshold-output-pres`
Status: `LEAN_FULL` on Mathlib $\mathbb{R}$ (`REAL_MATHLIB`).

## 9 Proof dependencies

- Absolute-value ball arithmetic only.
- No other operator theorems.

## 10 Consequences

Primitive scalar gate used by multi-threshold, abs-threshold, interval membership, and noisy threshold variants.